Juan David Muñoz-Bolaños

R&D Optical Engineer | Machine Vision & Metrology

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SUMMARY

R&D Optical Engineer based in **Tirol** with a dual background in **Industrial Machine Vision** and **High-Precision Microscopy**. Expert in designing optical inspection systems, implementing illumination solutions for defect inspection, and developing adaptive optics algorithms. Educated in **Germany** (M.Sc.) and **Austria** (Ph.D.). Proficient in transferring complex optical concepts—such as interferometry and wavefront shaping—into robust, high-speed hardware solutions suitable for semiconductor and industrial applications.

TECHNICAL SKILLS

Machine Vision & Inspection
MVTec HALCON, OCR (Optical Character Recognition), Laser Profilometry, Industrial Illumination Design, Camera Calibration, Defect Detection, Quality Control Systems.

Optical Engineering
Adaptive Optics (SLM, Texas Instruments DMD), Interferometry (Phase-stepping), Spectrometers, Raman Microscopy, Multiphoton Imaging, Beam Shaping, Lithography concepts.

Software & Computing
Python (NumPy, SciPy), LabVIEW (FPGA Real-time), GPU Computing (CuPy), MATLAB, Image Processing Algorithms.

Languages
English (Proficient), German (B2/Working Proficiency), Spanish (Native).

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher - Adaptive Optics & Metrology — Jan 2022 – Present

- **High-Precision Metrology:** Developed a setup using a novel transmissive Spatial Light Modulator (SLM) combined with phase-stepping interferometry to correct optical aberrations. Achieved high-resolution observation in two-photon microscopy (e.g., neuronal imaging).
- **Deep Signal Restoration:** Engineered a system using Raman microscopy and transmissive wavefront shaping to restore optical signals at significant penetration depths in scattering media.
- **High-Speed Hardware Control:** Currently implementing a **Texas Instruments DLP/DMD device** (reflection mode) to correct optical aberrations at high speeds, enabling real-time in-vivo image restoration.
- **Algorithm Development:** Accelerated holographic algorithms using GPU computing (CuPy) for rapid phase retrieval.

Leibniz Institute of Photonic Technology (IPHT) — Jena, Germany
Assistant Researcher — Jun 2020 – Dec 2021

- Assembled multimodal setups for vibrational imaging (Raman/IR) applied to rapid diagnostics.
- Integrated photothermal microscopy hardware for environmental microplastic detection.

INTECOL S.A.S. Medellin, Colombia
Machine Vision Engineer Mar 2018 – Mar 2019

- **Automated Quality Control (OCR):** Designed and deployed an inspection system for bottling lines. Implemented custom illumination and **HALCON** software algorithms to read and verify OCR codes on printed labels, ensuring 100% traceability.
- **Safety-Critical Metrology:** Developed a **laser profilometry system** to inspect the thickness of urban cable car lines (Metro de Medellín). Analyzed cable properties critical for public transportation safety.
- Managed full project lifecycles, from selecting industrial cameras/optics to on-site installation and calibration.

EDUCATION

Medical University of Innsbruck Innsbruck, Austria
Ph.D. Advanced Microscopy & Optics Expected 2026

Friedrich Schiller University Jena Jena, Germany
M.Sc. in Photonics 2019 – 2021

University of Cauca Colombia
B.Sc. in Physics Engineering 2012 – 2018

SELECTED PUBLICATIONS & AWARDS

- **J. D. Muñoz-Bolaños**, et al. "Scatter compensation for enabling deep two-photon microscopy using a MEMS phase light modulator," *Proc. SPIE* (2026).
- **J. D. Muñoz-Bolaños**, et al. "Wavefront corrections over large fields of view via beam cone tomography." *Proc. SPIE*, (2025).
- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).

REFERENCES

Prof. Monika Ritsch-Marte
Medical University of Innsbruck
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Prof. Jürgen Popp
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